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ALTERED ELEMENT CONCENTRATION IN SERUM, HEART, LIVER AND KIDNEY TISSUES AFTER EPINEPHRINE ADMINISTRATION IN RATS

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ABSTRACT

The effect of epinephrine administration (1mg/kg body weight) on trace elements and electrolytes levels of serum, liver, heart and kidney tissues was studied in rats. Epinephrine administration decreased sodium, potassium and calcium in heart, liver and kidney tissues. A significant elevation in copper and zinc level of serum, heart and kidney tissue was observed in epinephrine injected rats. Iron content was significantly decrease in serum and kidney tissue where as in live tissue significant elevation was observed after epinephrine administration. The results suggested that epinephrine administration causes alterations in element concentration both in serum and tissues. The results may help towards an understanding of the relationship between elevated catecholamine levels in emotional stress condition and sudden death through a process of altered elements concentration.

INTRODUCTION

Profound emotional stress such as joy, fear and anger can lead to sudden death probably as a result of high level of plasma catecholamines (Johanson, 1981). The mechanism of death is most likely hypokalemia induced ventricular fibrillation or ventricular tachycardia resulting in circulatory arrest. Previous studies shows that extrarenal tissue play an important role in acute potassium tolerance. Both insulin and adrenal hormones appear to play a role in the extrarenal potassium tolerance (Defronzo et al., 1981). The relationship of sodium, potassium, calcium and other elements to the process of ventricular tachycardia, fibrillation and hypertension has been established both in men and experimental animals and it is suggested that their imbalance could contribute to the appearance of long term variations in blood pressure (Mahboob et al., 1991; D'Silva, 1934; Stanton et al., 1987; McCarron, 1987). In order to investigate the involvement of elements in the process of sudden death due to high circulating catecholamine the present study was designed to study the effect of epinephrine administration on element concentration in serum, heart, liver and kidney tissue in rats.

MATERIALS AND METHODS

Locally bred male albino wistar rats weighing 180-250 grams body weight were used during the study. Animal were individually caged in a quiet temperature controlled room (23 ± 4°C). Rats had free access to water and standard diet. After two weeks of habituation rats were randomly divided into control and test groups (n = 16 in each). Test group received intraperitoneal injection of epinephrine 1.0 mg/kg body weight. Controls were treated identically with deionized water.

Sample collection:

After one hour of injections animals were decapitated and blood was collected. Heart, liver and kidney were excised, trimmed of connective tissues, rinsed with deionized water to eliminate blood contamination, dried with filter paper and weighed.

Sample preparation:

Serum and tissue samples were prepared according to the method described earlier (Mahboob et al., 1996). Simply heart, liver, and kidney were digested for 3 hours at room temperature and then at 70°C for another 3 hours with addition of about 300 mg tissues/ml nitric acid. The resulting clear solution was diluted with deionized water and the diluted samples were analyzed for sodium, potassium and calcium by flame photometry and iron, lead, copper and zinc by atomic absorption spectrophotometry (Clegg et al., 1982).

Statistical Analysis:

Results were expressed as mean ± S.D. Statistical significance of control and test values were evaluated by unpair student's t-test (Zar, 1974).

RESULTS AND DISCUSSION

Results are shown in Table I, II, III and IV. For most of the element studied our resulted confirmed that numerous modifications occurred in serum, and tissue content after epinephrine administration.

Concentration of sodium and potassium in serum, heart, liver and kidney were decreased. Although a slight elevated level of potassium was observed in heart tissue but the results were non significant (Table I-IV).

The epinephrine induced fall in serum potassium was demonstrated more than half a century ago (D'Silva, 1934). The mechanism of this decrease in serum potassium is a beta 2 adrenergic receptor stimulation of adenylate cyclase., which inturn stimulates the Na- K-ATPase in cell membrane (Classen and Flatman, 1980). According to the present results (Table I-IV), sodium level decrease in serum and all the tissues tested. This decrease of sodium by epinephrine administration may be due to increased prostaglandin synthesis in the renal tissue. Epinephrine infusion stimulate the excretion of prostaglandins (Loup et al., 1986). The increased prostaglandin after epinephrine administration causes an increased excretion of sodium in urine which results in a decreased level of sodium in serum as well in heart, liver and kidney tissues as shown in our results (Table I-IV).

Concentration of calcium was found to be decreased significantly in heart, liver and kidney tissues (Tables II, III, IV) whereas slight elevation in serum was observed which is nonsignificant (Table I). It is suggested that the decreased tissue content of calcium is due to mobilization in to the blood and then by increased excretion rate. The effect is also thought to be due to increased prostaglandin synthesis after epinephrine administration. Another possibility is epinephrine's activation of adenylatecyclase and thus the modification of membrane permeability with increasing Mg/Ca shift (Teo and Wang, 1973; Paul and Harvey, 1984) and in conclusion a decreased calcium content observed in tissue (Table II, III, IV).

Copper display a noticeable modification after epinephrine administration with a marked increase in serum (14.4%), heart (47.9%) and kidney (26.3%). Previous studies showed similar variations in hypertensive rats (Berthelots et al., 1987). The similarity shows a possible involvement of copper in blood pressure elevation after the administration of epinephrine. It has been observed that intravenous injection of copper is accompanied by a marked increase in peripheral vascular resistance in sheep (Ahmed et al., 1984) and a copper deficiency in animals underlies hypertension which may particularly result in disturbance in characteristics of structure and reactivity of the myocardium and vascular wall (Folkon et al., 1970).

A significant elevation in the zinc level of serum, heart and kidney tissue was observed in the present study (Table I, III, IV). The increase in serum and tissue Zn content could be related to abnormally high sympathetic neurone activity due to epinephrine administration (Judy et al., 1976; Baucher et al., 1984).

With regards to iron significant decrease in serum and kidney tissue observed under our experimental conditions (Table I, IV). Whereas in liver tissue significant elevation was observed after epinephrine administration (Table II). Possibly this decline of iron in serum and kidney tissue may be a response of epinephrine induced stress causing the mobilization of iron resulting in a decrease in serum level and increased level in liver tissue (Donnelly, 1978; Khan and Rahman, 1984).

The comparison of serum and tissue variations during the present study reveals considerable differences suggesting that it would be difficult to draw a conclusion on disturbances in elements metabolism using serum values only.

In conclusion the present study suggest that epinephrine administration causes alterations in element concentration both in serum and tissues and these variations in the concentration of elements in different tissues could be related in some way to the elevated blood pressure and heart rate in the process of sudden death during emotional stress when circulating serum catecholamine levels are raised.

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Table I
Effect of Intraperitoneal administration of epinephrine (1 mg/kg) on Serum concentration of elements in rats

Element	Control	Test	% Change
Na (mg/l)	106 ± 16.0	103 ± 12.0	2.79
K (mg/l)	3.05 ± 0.67	2.61 ± 0.45*	14.51
Ca (mg/l)	1.51 ± 0.65	1.69 ± 0.56	11.78
Cu (mg/l)	16.50 ± 1.25	18.92 ± 1.84	14.40
Zn (mg/l)	20.13 ± 1.98	26.83 ± 1.91	33.28
Pb (mg/l)	7.34 ± 5.04	7.14 ± 6.25	2.72
Fe (mg/l)	0.14 ± 0.01	0.09 ± 0.01*	35.71

Values are mean ± S.D. (in μg/g of tissue wet weight)
*$P < 0.001$, ** $P < 0.01$

Table II
Effect of Intraperitoneal administration of epinephrine (1 mg/kg) on element concentration in Liver tissue of rats

Element	Control	Test	% Change
Na	68.0 ± 4.0	50.0 ± 9.0*	26.47
K	89.0 ± 9.0	78.0 ± 10.0**	12.36
Ca	1.75 ± 0.3	1.17 ± 0.19*	33.14
Cu	3.71 ± 0.48	3.69 ± 1.69	0.53
Zn	24.40 ± 11.6	20.17 ± 3.53	17.33
Pb	2.65 ± 1.16	2.43 ± 0.73	6.92
Fe	116 ± 16.21	138.0 ± 23.2**	19.1

Values are mean ± S.D. (in μg/g of tissue wet weight)
*$P < 0.001$, ** $P < 0.01$

Table III
Effect of Intraperitoneal administration of epinephrine (1 mg/kg) on concentration of elements in Heart tissue of arts

Element	Control	Test	% Change
Na	64.0 ± 6.0	62.0 ± 7.0	3.12
K	66.0 ± 10.0	68.0 ± 7.0	3.03
Ca	1.41 ± 0.16	1.16 ± 0.1**	17.73
Cu	3.36 ± 1.16	4.97 ± 1.1*	47.91
Zn	19.51 ± 10.0	24.69 ± 7.73*	26.55
Pb	2.71 ± 0.44	2.60 ± 0.46	4.05
Fe	98.86 ± 8.84	102.58 ± 38.49	3.70

Values are mean ± S.D. (in μg/g of tissue wet weight)
*$P < 0.05$, **$P < 0.001$

Table IV
Effect of Intraperitoneal administration of epinephrine (1 $\mu g/kg$) on concentration of elements in Kidney tissue of rats.

Element	Control	Test	% Change
Na	83.0 ± 8.0	73.0 ± 9.0*	12.08
K	62.0 ± 5.0	55.0 ± 11.0***	11.29
Ca	2.09 ± 0.13	1.43 ± 0.19**	31.57
Cu	3.57 ± 0.70	4.51 ± 0.58*	26.33
Zn	18.27 ± 5.5	24.82 ± 3.34**	35.85
Pb	3.12 ± 0.69	3.27 ± 0.37	4.80
Fe	107 ± 40.89	96.0 ± 20.72***	10.03

Values are mean ± S.D. (in $\mu g/g$ of tissue wet weight)
*P < 0.001, **P < 0.01, *** P < 0.05

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